

FDU 回归分析 Homework 01

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A

Problem 1

Suppose the experimenter postulates a model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (i = 1, \dots, n)$$

where β_0 is known and $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$)

- (a) What is the appropriate least squares estimator of β_1 ? Justify your answer.
- (b) What is the variance of the estimator of β_1 in (a)?

Part (a)

What is the appropriate least squares estimator of β_1 ? Justify your answer.

Solution:

$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ ($i = 1, \dots, n$) 可以写成向量形式 $y = \beta_0 \mathbf{1}_n + \beta_1 x + \varepsilon$

残差平方和 $\text{RSS}(\beta_1) := \|y - \beta_0 \mathbf{1}_n - \beta_1 x\|_2^2$

参数 β_1 的最小均方估计量 $\hat{\beta}_1$ 是以下优化问题的最小值点:

$$\begin{aligned}\hat{\beta}_1 &:= \arg \min_{\beta_1 \in \mathbb{R}} \text{RSS}(\beta_1) \\ \nabla_{\beta_1} \text{RSS}(\beta_1) &= \nabla_{\beta_1} \{\|y - \beta_0 \mathbf{1}_n - \beta_1 x\|_2^2\} \\ &= -2x^T (y - \beta_0 \mathbf{1}_n - \beta_1 x) \\ \nabla_{\beta_1}^2 \text{RSS}(\beta_1) &= 2x^T x \geq 0\end{aligned}$$

这是一个凸优化问题, 因此最小值点 $\hat{\beta}_1$ 就是驻点.

令 $\nabla_{\beta_1} \text{RSS}(\beta_1) = -2x^T (y - \beta_0 \mathbf{1}_n - \beta_1 x) = 0$, 解得:

$$\hat{\beta}_1 = \frac{x^T (y - \beta_0 \mathbf{1}_n)}{x^T x} = \frac{\sum_{i=1}^n x_i (y_i - \beta_0)}{\sum_{i=1}^n x_i^2}$$

Part (b)

What is the variance of the estimator of β_1 in (a)?

Solution:

注意到 $\varepsilon \sim N(0_n, \sigma^2 I_n)$

$$\begin{aligned}\hat{\beta}_1 &= \frac{x^T (y - \beta_0 \mathbf{1}_n)}{x^T x} \\ &= \frac{x^T (\beta_0 \mathbf{1}_n + \beta_1 x + \varepsilon - \beta_0 \mathbf{1}_n)}{x^T x} \\ &= \beta_1 + \frac{x^T \varepsilon}{x^T x} \\ (\hat{\beta}_1 - \beta_1) &= \frac{x^T \varepsilon}{x^T x} \\ &\sim N\left(\frac{x^T}{x^T x} 0_n, \frac{x^T}{x^T x} \sigma^2 I_n \frac{x}{x^T x}\right) \\ &= N\left(0, \frac{\sigma^2}{x^T x}\right)\end{aligned}$$

因此 $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{x^T x})$, 表明 $\hat{\beta}_1$ 是 β_1 的无偏估计量, 且 $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{x^T x}$

没见过这种分割线的写法, 可以确认一下出处.

Problem 2

Consider a model $y_i = \beta + \varepsilon_i$ where β is a constant parameter and $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$)
Find LSE and MLE for β

Solution:

$y_i = \beta + \varepsilon_i$ ($i = 1, \dots, n$) 可以写成紧凑形式 $y = \beta 1_n + \varepsilon$

首先推导 β 的 LSE:

$$\hat{\beta}_{\text{LSE}} := \arg \min_{\beta \in \mathbb{R}} \|y - \beta 1_n\|_2^2$$

令 $\nabla_{\beta} \{\|y - \beta 1_n\|_2^2\} = -1_n^T \cdot 2(y - \beta 1_n) = 0$, 解得 $\hat{\beta}_{\text{LSE}} = \frac{1_n^T y}{1_n^T 1_n} = \frac{1}{n} 1_n^T y = \bar{y}$

其次推导 β 的 MLE:

注意到 $y = \beta 1_n + \varepsilon \sim N(\beta 1_n, \sigma^2 I_n)$, 故对数似然函数为:

$$\begin{aligned} L(\beta, \sigma^2) &:= \log\{P\{N(\beta 1_n, \sigma^2 I_n) = y\}\} \\ &= \log\left\{\frac{1}{\sqrt{(2\pi)^n \det(\sigma^2 I_n)}} \exp\left\{-\frac{1}{2}(y - \beta 1_n)^T (\sigma^2 I_n)^{-1} (y - \beta 1_n)\right\}\right\} \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|y - \beta 1_n\|_2^2 \end{aligned}$$

对于任意固定的 $\sigma^2 > 0$, 则我们有:

$$\begin{aligned} \hat{\beta}_{\text{MLE}} &:= \arg \max_{\beta \in \mathbb{R}} L(\beta, \sigma^2) \\ &= \arg \max_{\beta \in \mathbb{R}} \left\{-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|y - \beta 1_n\|_2^2\right\} \\ &= \arg \min_{\beta \in \mathbb{R}} \|y - \beta 1_n\|_2^2 \\ &= \hat{\beta}_{\text{LSE}} \\ &= \bar{y} \end{aligned}$$

固定 $\beta = \hat{\beta}_{\text{MLE}} = \bar{y}$, 则我们有:

$$\begin{aligned} \hat{\sigma}_{\text{MLE}}^2 &:= \arg \max_{\sigma^2 > 0} L(\bar{y}, \sigma^2) \\ &= \arg \max_{\sigma^2 > 0} \left\{-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|y - \bar{y} 1_n\|_2^2\right\} \\ &= \text{solution of } \left\{-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|y - \bar{y} 1_n\|_2^2 = 0\right\} \\ &= \frac{1}{n} \|y - \bar{y} 1_n\|_2^2 \end{aligned}$$

以. 3
这种写法...

Problem 3

A sample of n boys and n girls is taken from a school and their heights are measured.

Let $y_1, \dots, y_n$ denote the heights of the n girls, and $y_{n+1}, \dots, y_{2n}$ denote those of the n boys.

It is believed that $y_i = \alpha + \beta x_i + \varepsilon_i$ where $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, 2n$) and $x_i = \begin{cases} -1 & i = 1, \dots, n \\ +1 & i = n+1, \dots, 2n \end{cases}$

Find the least squares estimators of β

- Note: x is a dummy variable used to represent gender.

Solution:

我们记

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{2n} \end{bmatrix} = \begin{bmatrix} -1 \\ \vdots \\ -1 \\ +1 \\ \vdots \\ +1 \end{bmatrix} \quad X = [1_{2n}; x] \quad \gamma = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \\ \varepsilon_{n+1} \\ \vdots \\ \varepsilon_{2n} \end{bmatrix}$$

$$y^{(1)} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad y^{(2)} = \begin{bmatrix} y_{n+1} \\ \vdots \\ y_{2n} \end{bmatrix} \quad y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \end{bmatrix}$$

则我们可将 $y_i = \alpha + \beta x_i + \varepsilon_i$ ($i = 1, \dots, 2n$) 记为 $y = \alpha 1_{2n} + \beta x + \varepsilon = X\gamma + \varepsilon$

$$\begin{aligned} \hat{\gamma} &:= \arg \min_{\gamma \in \mathbb{R}^2} \text{RSS}(\gamma) \\ &= \arg \min_{\gamma \in \mathbb{R}^2} \|y - X\gamma\|_2^2 \\ &= \text{solution of } \nabla_{\gamma} \{\|y - X\gamma\|_2^2\} = -2X^T(y - X\gamma) = 0_2 \\ &= (X^T X)^{-1} X^T y \\ &= \begin{bmatrix} 1_{2n}^T 1_{2n} & 1_{2n}^T x \\ 1_{2n}^T x & x^T x \end{bmatrix}^{-1} \begin{bmatrix} 1_{2n}^T y \\ x^T y \end{bmatrix} \\ &= \begin{bmatrix} 2n & 0 \\ 0 & 2n \end{bmatrix}^{-1} \begin{bmatrix} \bar{y} \\ x^T y \end{bmatrix} \\ &= \frac{1}{2n} \begin{bmatrix} \bar{y} \\ -1_n^T y^{(1)} + 1_n^T y^{(2)} \end{bmatrix} \end{aligned}$$

因此 $\hat{\beta}$ 作为 $\hat{\gamma}$ 的第 2 个分量, 它为:

$$\hat{\beta} = \frac{1}{2n} (-1_n^T y^{(1)} + 1_n^T y^{(2)}) = \frac{1}{2n} \sum_{i=1}^n (y_{n+i} - y_i)$$

Problem 4

Consider a simple linear regression model:

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad (i = 1, \dots, n)$$

where $x_1, \dots, x_n$ are constants, α, β are parameters, $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$)

A weighted least square estimators for α, β are obtained by minimizing the Residual Sum of Squares:

$$\text{RSS}(\alpha, \beta) := \sum_{i=1}^n w_i (y_i - \alpha - \beta x_i)^2$$

where $w_1, \dots, w_n$ are some predefined known **positive** constant values and $\sum_{i=1}^n w_i = 1$

Find the weighted least square estimators for α, β and their covariance matrix.

Solution:

$y_i = \alpha + \beta x_i + \varepsilon_i$ ($i = 1, \dots, n$) 可以写成紧凑形式 $y = \alpha 1_n + \beta x + \varepsilon$, 其中 $\varepsilon \sim N(0_n, \sigma^2 I_n)$

记 $\begin{cases} W = \text{diag}\{w_1, \dots, w_n\} \\ w = [w_1, \dots, w_n]^T \\ \gamma = [\alpha, \beta]^T \\ X = [1_n; x] \end{cases}$ 则我们有: W = W 1_n

$$y = \alpha 1_n + \beta x + \varepsilon = X\gamma + \varepsilon$$

$$\text{RSS}(\alpha, \beta) := \sum_{i=1}^n w_i (y_i - \alpha - \beta x_i)^2 = (y - X\gamma)^T W (y - X\gamma) \text{ where } W \text{ satisfies } \begin{cases} W \succ 0 \\ \text{tr}(W) = 1 \end{cases}$$

于是 γ 的最小二乘估计量为如下优化问题的全局最小点:

$$\hat{\gamma} := \arg \min_{\gamma \in \mathbb{R}^2} (y - X\gamma)^T W (y - X\gamma)$$

$$\begin{aligned} \nabla_{\gamma} \{(y - X\gamma)^T W (y - X\gamma)\} &= \nabla_{\gamma} \{y^T W y - 2y^T W X \gamma + \gamma^T X^T W X \gamma\} \\ &= -2X^T W y + 2X^T W X \gamma \\ \nabla_{\gamma}^2 \{(y - X\gamma)^T W (y - X\gamma)\} &= 2X^T W X \succ 0 \end{aligned}$$

因此这是一个无约束凸优化问题，其全局最小点即为驻点。

令 $\nabla_{\gamma} \{(y - X\gamma)^T W (y - X\gamma)\} = -2X^T W y + 2X^T W X \gamma = 0_2$ 可得：

$$\begin{aligned} \hat{\gamma} &= (X^T W X)^{-1} X^T W y \\ &= \begin{bmatrix} 1_n^T W 1_n & 1_n^T W x \\ x^T W 1_n & x^T W x \end{bmatrix}^{-1} \begin{bmatrix} 1_n^T W y \\ x^T W y \end{bmatrix} \quad (\text{note that } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ if } ad - bc \neq 0) \\ &= \frac{1}{(1_n^T W 1_n)(x^T W x) - (1_n^T W x)(x^T W 1_n)} \begin{bmatrix} x^T W x & -1_n^T W x \\ -x^T W 1_n & 1_n^T W 1_n \end{bmatrix} \begin{bmatrix} 1_n^T W y \\ x^T W y \end{bmatrix} \quad (\text{note that } \begin{cases} 1_n^T W 1_n = \sum_{i=1}^n w_i = 1 \\ W 1_n = w \end{cases}) \\ &= \frac{1}{x^T W x - (w^T x)^2} \begin{bmatrix} (x^T W x)(w^T y) - (w^T x)(x^T W y) \\ -(w^T x)(w^T y) + x^T W y \end{bmatrix} \end{aligned}$$

(根据上述讨论我们也知道 note that $(X^T W X)^{-1} = \frac{1}{x^T W x - (w^T x)^2} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix}$)

因此 α, β 的加权最小二乘估计量 $\hat{\alpha}, \hat{\beta}$ 分别为 $\hat{\gamma}$ 的两个分量：

$$\begin{aligned} \hat{\alpha} &= \frac{(x^T W x)(w^T y) - (w^T x)(x^T W y)}{x^T W x - (w^T x)^2} \\ &= \frac{(\sum_{i=1}^n w_i x_i^2)(\sum_{i=1}^n w_i y_i) - (\sum_{i=1}^n w_i x_i)(\sum_{i=1}^n w_i x_i y_i)}{\sum_{i=1}^n w_i x_i^2 - (\sum_{i=1}^n w_i x_i)^2} \\ \hat{\beta} &= \frac{x^T W y - (w^T x)(w^T y)}{x^T W x - (w^T x)^2} \\ &= \frac{\sum_{i=1}^n w_i x_i y_i - (\sum_{i=1}^n w_i x_i)(\sum_{i=1}^n w_i y_i)}{\sum_{i=1}^n w_i x_i^2 - (\sum_{i=1}^n w_i x_i)^2} \end{aligned}$$

下面我们考虑 $\hat{\gamma} = [\hat{\alpha}, \hat{\beta}]^T$ 的分布：

$$\begin{aligned} \hat{\gamma} &= (X^T W X)^{-1} X^T W y \\ &= (X^T W X)^{-1} X^T W (X\gamma + \varepsilon) \\ &= \gamma + (X^T W X)^{-1} X^T W \varepsilon \\ \hat{\gamma} - \gamma &= (X^T W X)^{-1} X^T W \varepsilon \\ &\sim N((X^T W X)^{-1} X^T W \cdot 0_n, (X^T W X)^{-1} X^T W \cdot \sigma^2 I_n \cdot [(X^T W X)^{-1} X^T W]^T) \\ &= N(0_n, \sigma^2 \cdot (X^T W X)^{-1} X^T W^2 X (X^T W X)^{-1}) \end{aligned}$$

因此 $\hat{\alpha}, \hat{\beta}$ 的协方差矩阵为：

$$\begin{aligned} \text{Cov}(\hat{\gamma}) &= \sigma^2 \cdot (X^T W X)^{-1} X^T W^2 X (X^T W X)^{-1} \quad (\text{note that } (X^T W X)^{-1} = \frac{1}{x^T W x - (w^T x)^2} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix}) \\ &= \frac{\sigma^2}{[x^T W x - (w^T x)^2]^2} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix} \begin{bmatrix} 1_n^T W^2 1_n & 1_n^T W^2 x \\ x^T W^2 1_n & x^T W^2 x \end{bmatrix} \begin{bmatrix} x^T W x & -w^T x \\ -w^T x & 1 \end{bmatrix} \end{aligned}$$

其中 $\begin{cases} W = \text{diag}\{w_1, \dots, w_n\} \\ w = [w_1, \dots, w_n]^T \end{cases}$ (不想继续化简了，这没有什么意义)

这就对了，不解答没必要。
是漂亮的结果！

Problem 5

Consider a simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (i = 1, \dots, n)$$

where $x_1, \dots, x_n$ are constants, α, β are parameters, $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$)

Suppose $n = 3$ and three data points: $(-1, y_1), (0, y_2), (1, y_3)$

- (a) Find the design matrix X
- (b) Express the least squares estimators of β_0, β_1 in terms of y_1, y_2, y_3

- (c) Find out the expression of the regression sum of squares in terms of y_1, y_2, y_3
- (d) Suppose $y_3 = y_1 + 2$ and the total sum of squares $SST := S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = 2.5$
What is the value of the coefficient of determination R^2 ?
Does the regression line fit the three data points well?
- (e) Under the condition of (d), find the 95% confidence intervals for β_0, β_1 respectively.

Part (a)

Find the design matrix X

Solution:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

于是 $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ ($i = 1, 2, 3$) 可以写成紧凑形式 $y = X\beta + \varepsilon$ (其中 $\beta = [\beta_0, \beta_1]^T$)

Part (b)

Express the least squares estimators of β_0, β_1 in terms of y_1, y_2, y_3

Solution:

$$\begin{aligned} \hat{\beta} &:= \arg \min_{\beta \in \mathbb{R}^2} \|y - X\beta\|_2^2 \\ &= \text{solution of } \{-2X^T(y - X\beta) = 0_2\} \\ &= (X^T X)^{-1} X^T y \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3}(y_1 + y_2 + y_3) \\ \frac{1}{2}(y_3 - y_1) \end{bmatrix} \end{aligned}$$

因此我们有:

$$\begin{cases} \hat{\beta}_0 = \frac{1}{3}(y_1 + y_2 + y_3) = \bar{y} \\ \hat{\beta}_1 = \frac{1}{2}(y_3 - y_1) \end{cases}$$

Part (c)

Find out the expression of the regression sum of squares in terms of y_1, y_2, y_3

Solution:

$$\begin{aligned}
SSR &= \|\hat{y} - \bar{y} \mathbf{1}_3\|_2^2 \\
&= \|X\hat{\beta} - \frac{1}{3} \mathbf{1}_3^T y \mathbf{1}_3\| \quad (\text{note that } \hat{\beta} = (X^T X)^{-1} X^T y) \\
&= \|[X(X^T X)^{-1} X^T - \frac{1}{3} \mathbf{1}_3 \mathbf{1}_3^T] y\|_2^2 \\
&= y^T [X(X^T X)^{-1} X^T - \frac{1}{3} \mathbf{1}_3 \mathbf{1}_3^T] y \\
&= y^T \left(\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}^T - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \left(\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \\ & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \left(\begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{6} \\ 0 & 0 & 0 \\ -\frac{1}{6} & 0 & \frac{1}{2} \end{bmatrix} y \\
&= \frac{1}{2} (y_1 - y_3)^2
\end{aligned}$$

Part (d)

Suppose $y_3 = y_1 + 2$ and the total sum of squares $SST := S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = 2.5$

What is the value of the coefficient of determination R^2 ?

Does the regression line fit the three data points well?

Solution:

由于 $y_3 = y_1 + 2$, 故 $SSR = \frac{1}{2} (y_1 - y_3)^2 = \frac{1}{2} \cdot 2^2 = 2$

因此 $R^2 = \frac{SSR}{SST} = \frac{2}{2.5} = 0.8$

这个数值很接近于 1, 故模型较好地拟合了 3 个数据点.

Part (e)

Under the condition of (d), find the 95% confidence intervals for β_0, β_1 respectively.

Solution:

根据 $\begin{cases} y_3 = y_1 + 2 \\ S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = y_1^2 + y_2^2 + y_3^2 - 3\bar{y}^2 = 2.5 \text{ 我们可以得到:} \\ \bar{y} = \frac{1}{3} (y_1 + y_2 + y_3) \end{cases}$

$$\begin{aligned}
2.5 &= y_1^2 + y_2^2 + (y_1 + 2)^2 - 3 \cdot \frac{1}{9} (2y_1 + y_2 + 2)^2 \\
&= 2y_1^2 + y_2^2 + 4y_1 + 4 - \frac{1}{3} (4y_1^2 + y_2^2 + 4 + 4y_1y_2 + 8y_1 + 4y_2) \\
&= \frac{2}{3} y_1^2 + \frac{2}{3} y_2^2 + \frac{4}{3} y_1 - \frac{4}{3} y_2 - \frac{4}{3} y_1y_2 + \frac{8}{3} \\
&= \frac{2}{3} (y_1^2 + y_2^2 + 1 + 2y_1 - 2y_2 - 2y_1y_2) + 2 \\
&= \frac{2}{3} (y_1 - y_2 + 1)^2 + 2
\end{aligned}$$

$$(y_1 - y_2 + 1)^2 = \frac{3}{4} \Rightarrow (y_1 - y_2) = -1 \pm \frac{\sqrt{3}}{2}$$

但遗憾的是, 根据已有数据我们仍然无法计算出 $\beta_0 = \bar{y}$, 我们就当 $\bar{y}$ 是已知的就好了.

根据 **Part (b)** 我们有:

$$\begin{aligned}
\hat{\beta} &:= \arg \min_{\beta \in \mathbb{R}^2} \|y - X\beta\|_2^2 \\
&= \text{solution of } \{-2X^T(y - X\beta) = 0_2\} \\
&= (X^T X)^{-1} X^T y \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{3}(y_1 + y_2 + y_3) \\ \frac{1}{2}(y_3 - y_1) \end{bmatrix} \quad (\text{note that } y_3 = y_1 + 2) \\
&= \begin{bmatrix} \bar{y} \\ 1 \end{bmatrix}
\end{aligned}$$

因此我们有:

$$\begin{aligned}
\hat{\beta} &= (X^T X)^{-1} X^T y \\
&= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\
&= \beta + (X^T X)^{-1} X^T \varepsilon \\
\hline
\hat{\beta} - \beta &= (X^T X)^{-1} X^T \varepsilon \\
&\sim N((X^T X)^{-1} X^T 0_n, (X^T X)^{-1} X^T \cdot \sigma^2 I_n \cdot [(X^T X)^{-1} X^T]^T) \\
&= N(0_2, \sigma^2 (X^T X)^{-1}) \\
&= N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1}\right) \\
&= N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{3}\sigma^2 & 0 \\ 0 & \frac{1}{2}\sigma^2 \end{bmatrix}\right)
\end{aligned}$$

而 σ^2 的无偏估计量 $s^2 = \frac{1}{n-2} \text{SSE} = \frac{1}{n-2} (\text{SST} - \text{SSR}) = \frac{1}{3-2} (2.5 - 2) = 0.5 \sim \sigma^2 \chi_{(1)}^2$

因此用于对 β_0, β_1 进行区间估计的枢轴量为:

$$\begin{aligned}
\frac{\hat{\beta}_0 - \beta_0}{\sqrt{\frac{1}{3}s^2}} &= \sqrt{6}(\bar{y} - \beta_0) \sim t_1 \\
\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{1}{2}s^2}} &= 2(1 - \beta_1) \sim t_1
\end{aligned}$$

因此 β_0 的 95% 置信区间为 $[\bar{y} - \frac{\sqrt{6}}{6} t_{1,0.975}, \bar{y} + \frac{\sqrt{6}}{6} t_{1,0.975}]$ (其中我们假设 $\bar{y}$ 已知)

而 β_1 的 95% 置信区间为 $[1 - \frac{1}{2} t_{1,0.975}, 1 + \frac{1}{2} t_{1,0.975}]$

其中 $t_{1,0.975} \approx 12.71$ 为自由度为 1 的 t 分布的 $1 - \frac{1}{2}(1 - 0.95) = 0.975$ 分位数.